Course Project MATH 7/8680

Saiqua Islam[1], Sajal Shovon[2], Haiyang Zhang[3]

[1]Herff College of Engineering, University of Memphis
[2]College of Arts and Sciences, University of Memphis
[3]College of Graduate Health Sciences, University of Tennessee Health Science Center

1 Introduction

1. Let $\{Y_1, \ldots, Y_N\}$ be independent Bernoulli random variables such that

$$Y_i \sim Ber(\Phi(\boldsymbol{x}_i'\boldsymbol{\beta}))$$

2. For the binary observations Y_i, introduce continuous latent variable $Z_i, i = 1, \ldots N$ such that

$$Y_i = 1 \text{ iff } Z_i > 0,$$

$$Y_i = 0 \text{ iff } Z_i \leq 0,$$

where, $Z_i = \boldsymbol{x}_i'\boldsymbol{\beta} + \epsilon_i$ and $\epsilon_i \sim N(0, \sigma^2)$.

3. **Priors**: Let $\tau(\boldsymbol{\beta}, \sigma^2) \propto \frac{1}{\sigma^2}$.

4. Using procedures developed Bayesian Linear Model in combination with that for Bayesian Analysis of Binary Responses,

   - find the joint posterior density of $\boldsymbol{\beta}, \sigma^2, Z_1, \ldots, Z_N$,
   - find the fully conditional posterior density of $, \boldsymbol{\beta}, \sigma^2, Z_1, \ldots, Z_N$
   - implement a Gibbs sampling to collect samples of $\boldsymbol{\beta}, \sigma^2, Z_1, \ldots, Z_N$.

5. **Validation**

   - Simulate data using known values of $\beta_0, \beta_1, \sigma^2$ and selected values $x_1, \ldots, x_N$ with

$$\boldsymbol{x}_i'\boldsymbol{\beta} = \beta_0 + \beta_1 x_i,$$

   - Sample $Z_i \sim \text{Ber}(N(\beta_0 + \beta_1 x_i, \sigma^2))$.
   - Use $N = 10, 15, 20, 30, 40$.

6. Write a fully documented R code for this project.

7. For $N = 10$, estimate $E(P(\beta_1 > 0 | Y_1 = \cdots = Y_7 = 1, Y_8 = \cdots = Y_{10} = 0)$ and intepreté this result.

2 Prior

Using non-informative **Jeffrey's Prior**:

$$\tau(\boldsymbol{\beta}, \sigma^2) \propto \frac{1}{\sigma^2}.$$

3 Joint Posterior and Fully Conditional Posteriors

Using procedures from the Bayesian linear model and the binary-response latent-variable formulation, we derive the joint posterior distribution of $(\boldsymbol{\beta}, \sigma^2, Z_1, \ldots, Z_N)$.

3.1 Joint Posterior

$$\begin{aligned}
\tau(\boldsymbol{\beta}, Z, \sigma^2 \mid y) &\propto f(y \mid Z, \boldsymbol{\beta}, \sigma^2)\, \tau(Z, \boldsymbol{\beta}, \sigma^2) \\
&\propto f(y \mid Z)\, \tau(Z, \boldsymbol{\beta}, \sigma^2) \qquad (y \text{ is completely determined by } Z) \\
&\propto f(y \mid Z)\, \tau(Z \mid \boldsymbol{\beta}, \sigma^2)\, \tau(\boldsymbol{\beta}, \sigma^2) \\
&\propto \prod_{i=1}^{N} \Big[I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0) \Big] \tau(Z \mid \boldsymbol{\beta}, \sigma^2)\, \tau(\boldsymbol{\beta}, \sigma^2) \\
&\propto \prod_{i=1}^{N} \Big[I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0) \Big] \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(Z_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{2\sigma^2} \right] \frac{1}{\sigma^2}
\end{aligned}$$

3.2 Fully Conditional Posteriors

Collecting terms accordingly from the joint posterior distribution provided above, we can get the fully conditional posteriors.

3.2.1 Fully conditional for σ^2

$$\begin{aligned}
\tau(\sigma^2 \mid \boldsymbol{\beta}, Z, y) &\propto (\sigma^2)^{-N/2} \exp\left[-\frac{\sum_{i=1}^{N} (Z_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{2\sigma^2} \right] (\sigma^2)^{-1} \\
&\propto (\sigma^2)^{-(N/2+1)} \exp\left[-\frac{\sum_{i=1}^{N} (Z_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{2\sigma^2} \right].
\end{aligned}$$

Thus,

$$\sigma^2 \mid \boldsymbol{\beta}, Z, y \sim IG\left(\frac{N}{2},\ \frac{(\mathbf{Z}-\mathbf{X}\boldsymbol{\beta})'(\mathbf{Z}-\mathbf{X}\boldsymbol{\beta})}{2}\right)$$

3.2.2 Fully conditional for $\boldsymbol{\beta}$

$$\begin{aligned}
\tau(\boldsymbol{\beta} \mid \sigma^2, Z, y) &\propto \exp\left[-\frac{(\mathbf{Z}-\mathbf{X}\boldsymbol{\beta})'(\mathbf{Z}-\mathbf{X}\boldsymbol{\beta})}{2\sigma^2}\right] \\
&\propto \exp\left[-\frac{(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}})'\mathbf{X}'\mathbf{X}(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}})}{2\sigma^2}\right] \\
&\propto \exp\left\{-\frac{(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}})'[\sigma^2(\mathbf{X}'\mathbf{X})^{-1}]^{-1}(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}})}{2}\right\},
\end{aligned}$$

where $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}$, from Bayesian Linear Model.

Thus,

$$\boldsymbol{\beta} \mid \sigma^2, Z, y \sim N\big((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\big)\,.$$

3.2.3 Fully conditional for Z_i

$$\tau(Z_i \mid \boldsymbol{\beta}, \sigma^2, y) \propto \Big[I(Z_i > 0, y_i = 1) + I(Z_i \le 0, y_i = 0)\Big]\ \exp\left[-\frac{(Z_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{2\sigma^2}\right].$$

Thus,

$$Z_i \mid \boldsymbol{\beta}, \sigma^2, y \sim N(\mathbf{x}_i'\boldsymbol{\beta}, \sigma^2) \quad \text{truncated to} \begin{cases} (0, \infty), & y_i = 1, \\ (-\infty, 0], & y_i = 0. \end{cases}$$

4 Parameter Identifiability

Under this model, there is parameter identifiability issue involved. We can look at the probability of observing $Y_i = 1 \mid \boldsymbol{\beta}, \sigma^2$ and $Y_i = 0 \mid \boldsymbol{\beta}, \sigma^2$ separately.

4.1 Probability that $Y_i = 1$

$$
\begin{aligned}
P(Y_i = 1 \mid \boldsymbol{\beta}, \sigma^2) &= P(Z_i > 0 \mid \boldsymbol{\beta}, \sigma^2) \\
&= P(\mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i > 0) \\
&= P(\varepsilon_i > -\mathbf{x}_i'\boldsymbol{\beta}) \\
&= P\left(\frac{\varepsilon_i}{\sigma} > -\frac{\mathbf{x}_i'\boldsymbol{\beta}}{\sigma}\right) \quad (\frac{\varepsilon_i}{\sigma} \sim N(0,1)) \\
&= 1 - \Phi\left(-\frac{\mathbf{x}_i'\boldsymbol{\beta}}{\sigma}\right) \quad \text{(Using standard normal CDF } \Phi) \\
&= \Phi\left(\frac{\mathbf{x}_i'\boldsymbol{\beta}}{\sigma}\right) \quad \text{(By symmetry: } \Phi(-a) = 1 - \Phi(a))
\end{aligned}
$$

4.2 Probability that $Y_i = 0$

Similarly,

$$P(Y_i = 0 \mid \boldsymbol{\beta}, \sigma^2) = 1 - \Phi\left(\frac{\mathbf{x}_i'\boldsymbol{\beta}}{\sigma}\right) = \Phi\left(-\frac{\mathbf{x}_i'\boldsymbol{\beta}}{\sigma}\right)$$

We notice that the probability of the binary response under this model, is determined by $\frac{\boldsymbol{\beta}}{\sigma}$ and **NOT** the absolute value of $\boldsymbol{\beta}$ or σ. The (misspecified) model actually becomes,

$$\boxed{Y_i \sim Ber(\Phi(\frac{\mathbf{x}_i'\boldsymbol{\beta}}{\sigma}))}$$

.

5 Data for Validation

The dataset used to validate the Gibbs sampling procedure comes from an insecticide dose–response study. In this experiment, insects were exposed to different dose levels of an insecticide, and the adverse response was recorded at each dose. For each dose level d_i, the number of insects exposed (N_i) and the number adversely affected (A_i) were observed. Table 5-1 summarizes the data provided. The data is given binomial instead of Bernoulli. Therefore, to transform the data into manageable form, take Dose=1.69 for example, there will be 6 **X=1.69** with 5 **Y=0** and 1 **Y=1**. Similarly, we transform the other dose-response information and get Table 5-2 for running Gibbs Sampler.

Table 5-1 Grouped data

Dose d_i	N_i (Exposed)	A_i
1.69	6	1
1.72	6	3
1.76	6	2
1.78	6	3
1.81	6	5
1.84	6	4
1.86	6	6
1.88	6	6

Table 5-2 Individual data

(Dose d_i, Response Y_i) Pairs				
(1.69, 0)	(1.69, 0)	(1.69, 0)	(1.69, 0)	(1.69, 0)
(1.69, 1)	(1.72, 0)	(1.72, 0)	(1.72, 0)	(1.72, 1)
(1.72, 1)	(1.72, 1)	(1.76, 0)	(1.76, 0)	(1.76, 0)
(1.76, 0)	(1.76, 1)	(1.76, 1)	(1.78, 0)	(1.78, 0)
(1.78, 0)	(1.78, 1)	(1.78, 1)	(1.78, 1)	(1.81, 0)
(1.81, 1)	(1.81, 1)	(1.81, 1)	(1.81, 1)	(1.81, 1)
(1.84, 0)	(1.84, 0)	(1.84, 1)	(1.84, 1)	(1.84, 1)
(1.84, 1)	(1.86, 1)	(1.86, 1)	(1.86, 1)	(1.86, 1)
(1.86, 1)	(1.86, 1)	(1.88, 1)	(1.88, 1)	(1.88, 1)
(1.88, 1)	(1.88, 1)	(1.88, 1)	(1.88, 1)	(1.88, 1)

6 R Document

The R document describing the procedure for Gibbs sampling in this case is given below. The **X** and **Y** are provided and described in Section 5.

```
library(mvtnorm)

set.seed(1)

X=c(rep(1.69, 6),
    rep(1.72, 6),
    rep(1.76, 6),
    rep(1.78, 6),
    rep(1.81, 6),
    rep(1.84, 6),
    rep(1.86, 6),
    rep(1.88, 6))
X=cbind(1, as.vector(X))
#X=cbind(1, as.vector(log(X, base=exp(1))))
Y=c(0,0,0,0,0,1,
    0,0,0,1,1,1,
    0,0,0,0,1,1,
    0,0,0,1,1,1,
    0,1,1,1,1,1,
    0,0,1,1,1,1,
    1,1,1,1,1,1,
    1,1,1,1,1,1)

N=length(Y)
p=ncol(X)

```

```r
n_iter=20000
burn_in=5000
thin=10

beta_samples=matrix(0, n_iter, p)
sigma2_samples=rep(0, n_iter)
Z_samples=matrix(0, n_iter, N)
theta_samples=matrix(0, n_iter, p)

inv_XtX=solve(t(X)%*%X)

# Initial values
beta=c(-0.5, 1)
sigma2=1

Z=numeric(N)

for (i in 1:n_iter) {

  # Sample Z_{i}
  for (j in 1:N) {
    flag=FALSE
    while (!flag) {
      z_candidate=rnorm(1, mean=X[j,]%*%beta, sd=sqrt(sigma2))
      if ((Y[j]==1 && z_candidate>0)||(Y[j]==0 && z_candidate<=0)) {
        Z[j]=z_candidate
        flag=TRUE
      }
    }
  }
  Z_samples[i,]=Z

  # Sample beta
  beta=as.vector(rmvnorm(1, mean=inv_XtX%*%t(X)%*%Z, sigma=sigma2*inv_XtX))
  beta_samples[i,]=beta

  # Sample sigma2
  sigma2=1/rgamma(1, shape=N/2, rate=t(Z-X%*%beta)%*%(Z-X%*%beta)/2)
  sigma2_samples[i]=sigma2

  # Store theta
  theta=beta/sqrt(sigma2)
  theta_samples[i,]=theta
}

# Remove burn-in and apply thinning
keep=seq(burn_in+1, n_iter, by=thin)
Z_post=Z_samples[keep,]
beta_post=beta_samples[keep,]
```

```r
sigma2_post=sigma2_samples[keep]
theta_post=theta_samples[keep,]

# Diagnostics

par(mfrow=c(2, 3), mar=c(4, 4, 2, 1))

plot(theta_post[,1], type='l',
     xlab = "Iteration", ylab = expression(beta[0]/sigma),
     main = expression("Trace Plot:" ~ beta[0]/sigma))
hist(theta_post[,1], breaks = 25, prob = TRUE,
     xlab = expression(beta[0]/sigma), main = expression("Posterior:" ~ beta[0]/sigma))
lines(density(theta_post[,1]))
acf(theta_post[,1], main = "")
title(main = expression("ACF:" ~ beta[0]/sigma))


plot(theta_post[,2], type='l',
     xlab = "Iteration", ylab = expression(beta[1]/sigma),
     main = expression("Trace Plot:" ~ beta[1]/sigma))
hist(theta_post[,2], breaks = 25, prob = TRUE,
     xlab = expression(beta[1]/sigma), main = expression("Posterior:" ~ beta[1]/sigma))
lines(density(theta_post[,2]))
acf(theta_post[,2], main = "")
title(main = expression("ACF:" ~ beta[1]/sigma))

# Sample statistics
cat("Posterior mean of theta_0:", round(mean(theta_post[,1]), 4), "\n")
cat("Posterior standard deviation of theta_0:", round(sd(theta_post[,1]), 4), "\n")
cat("95% credible interval for theta_0: [",
    round(quantile(theta_post[,1], 0.025), 4), ", ",
    round(quantile(theta_post[,1], 0.975), 4), "]\n\n")
cat("Posterior mean of theta_1:", round(mean(theta_post[,2]), 4), "\n")
cat("Posterior standard deviation of theta_1:", round(sd(theta_post[,2]), 4), "\n")
cat("95% credible interval for theta_1: [",
    round(quantile(theta_post[,2], 0.025), 4), ", ",
    round(quantile(theta_post[,2], 0.975), 4), "]\n")
```

7 Results and Diagnostics

Running the above R code, the results will show reasonable trace plots and posterior density plots for the identifiable parameters $\frac{\beta}{\sigma}$. The trace plots of the identifiable parameters will stabilize around the true posterior mean and the density plots are shown in Figure 7-1. And we can check the ACF

to see if there is any autocorrelation worth noting, also shown in Figure 7-1.

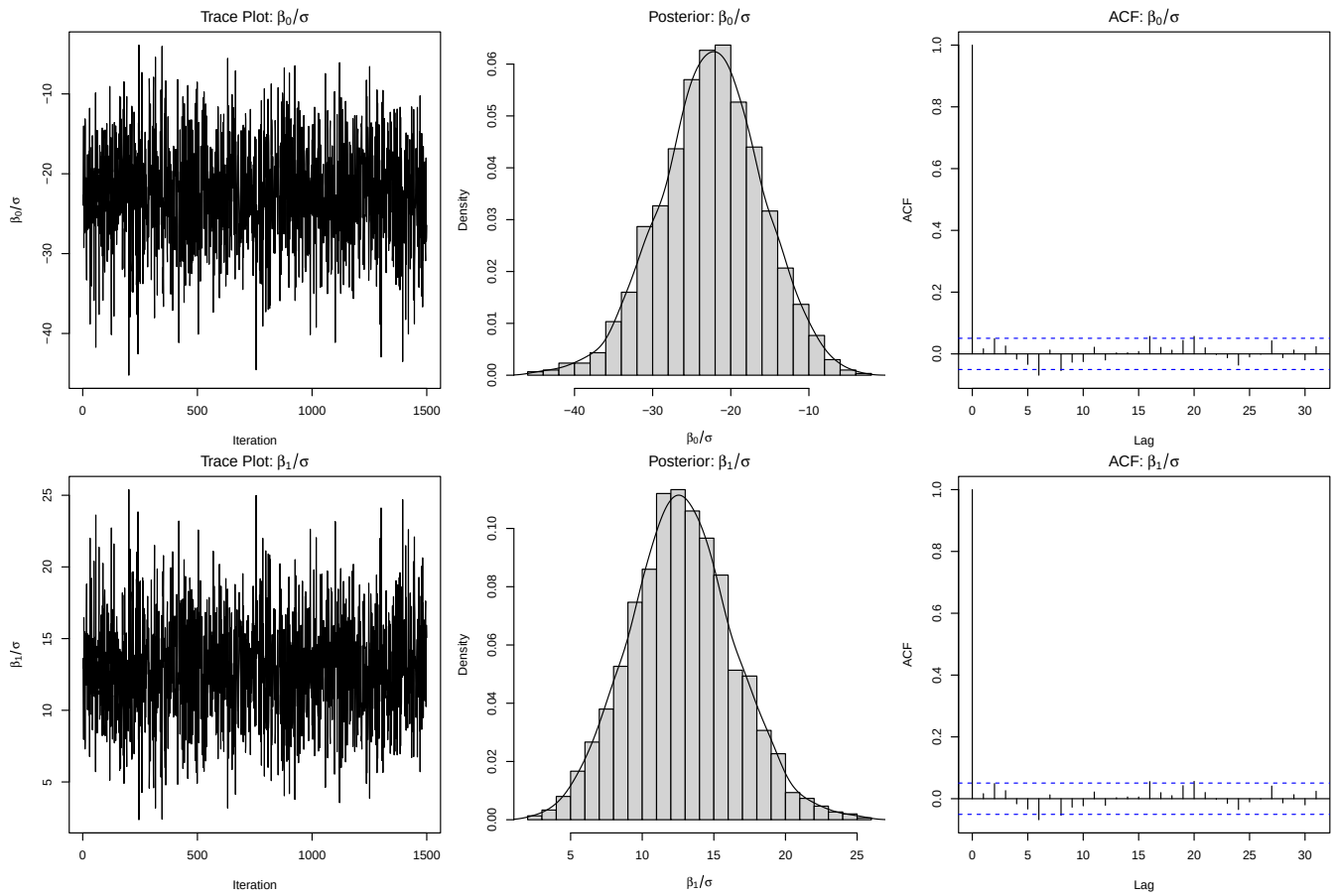

Figure 7-1 Trace plots, posterior density and ACF plots.

8 Hierarchical Analysis

Given the non-identifiability, we can reformulate the question into a hierarchical model, where we add an informative prior on the parameter $\boldsymbol{\beta}$.

$$
\begin{aligned}
Y_i = 1 \ \text{ iff } \ Z_i > 0 \\
Y_i = 0 \ \text{ iff } \ Z_i \leq 0 \\
z_i \mid \boldsymbol{\beta} \sim N(\mathbf{x}_i'\boldsymbol{\beta}, 1) \\
\boldsymbol{\beta} \mid \boldsymbol{\beta}_0, \sigma^2 \sim N(\mathbf{A}\boldsymbol{\beta}_0, \sigma^2\mathbf{I}_2) \\
\boldsymbol{\beta}_0 \propto 1
\end{aligned}
$$

$$\sigma^2 \propto \frac{1}{\sigma^2} \quad \text{(Jeffrey's prior)}$$

8.1 Joint Posterior

We can derive the joint posterior in this case,

$$\begin{aligned}
\tau(\boldsymbol{\beta}, \boldsymbol{\beta}_0, \sigma^2, z \mid y) &\propto f(y \mid Z, \boldsymbol{\beta}, \boldsymbol{\beta}_0, \sigma^2)\, \tau(Z, \boldsymbol{\beta}, \boldsymbol{\beta}_0, \sigma^2) \\
&\propto f(y \mid Z)\, \tau(Z, \boldsymbol{\beta}, \boldsymbol{\beta}_0, \sigma^2) \quad (y \text{ is completely determined by } Z) \\
&\propto f(y \mid Z) \tau(Z \mid \boldsymbol{\beta}) \tau(\boldsymbol{\beta} \mid \boldsymbol{\beta}_0, \sigma^2) \tau(\boldsymbol{\beta}_0, \sigma^2) \\
&\propto f(y \mid Z) \tau(Z \mid \boldsymbol{\beta}) \tau(\boldsymbol{\beta} \mid \boldsymbol{\beta}_0, \sigma^2) \tau(\boldsymbol{\beta}_0) \tau(\sigma^2) \quad \text{(independent priors)} \\
&\propto \prod_{i=1}^{N} \Big[I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0) \Big] \tau(Z \mid \boldsymbol{\beta}) \tau(\boldsymbol{\beta} \mid \boldsymbol{\beta}_0, \sigma^2) \tau(\boldsymbol{\beta}_0) \tau(\sigma^2) \\
&\propto \prod_{i=1}^{N} \Big[I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0) \Big] \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(Z_i - \mathbf{x}_i' \boldsymbol{\beta})^2}{2} \right] \\
&\quad \tau(\boldsymbol{\beta} \mid \boldsymbol{\beta}_0, \sigma^2) \tau(\boldsymbol{\beta}_0) \tau(\sigma^2) \\
&\propto \prod_{i=1}^{N} \Big[I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0) \Big] \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(Z_i - \mathbf{x}_i' \boldsymbol{\beta})^2}{2} \right] \\
&\quad \frac{1}{(2\pi)^{2/2} |\sigma^2 \mathbf{I}_2|^{1/2}} \exp\left[-\frac{1}{2} (\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)' (\sigma^2 \mathbf{I}_2)^{-1} (\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0) \right] \frac{1}{\sigma^2} \\
&\propto \prod_{i=1}^{N} \Big[I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0) \Big] \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(Z_i - \mathbf{x}_i' \boldsymbol{\beta})^2}{2} \right] \\
&\quad \frac{1}{(2\pi\sigma^2)^{2/2}} \exp\left[-\frac{1}{2\sigma^2} (\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)' (\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0) \right] \frac{1}{\sigma^2} \\
&\propto \prod_{i=1}^{N} \big[I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0) \big] \\
&\quad \exp\left[-\frac{1}{2} \sum_{i=1}^{N} (Z_i - \mathbf{x}_i' \boldsymbol{\beta})^2 - \frac{1}{2\sigma^2} (\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)' (\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0) \right] (\sigma^2)^{-(2/2+1)}
\end{aligned}$$

8.2 Fully Conditional Posteriors

Collecting terms accordingly from the joint posterior distribution provided above, we can get the fully conditional posteriors.

8.2.1 Fully conditional for σ^2

$$\tau(\sigma^2 \mid \boldsymbol{\beta}, \boldsymbol{\beta}_0, Z, y) \propto (\sigma^2)^{-(2/2+1)} \exp\left[-\frac{(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)'(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)}{2\sigma^2}\right]$$

Thus,

$$\boxed{\sigma^2 \mid \boldsymbol{\beta}, \boldsymbol{\beta}_0, y \sim IG\left(1, \frac{(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)'(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)}{2}\right) \quad (\sigma^2 \text{ is independent of } Z)}$$

8.2.2 Fully conditional for $\boldsymbol{\beta}$

$$\begin{aligned}
\tau(\boldsymbol{\beta} \mid \boldsymbol{\beta}_0, \sigma^2, Z, y) &\propto \exp\left[-\frac{1}{2}\sum_{i=1}^{N}(Z_i - \mathbf{x}_i'\boldsymbol{\beta})^2 - \frac{1}{2\sigma^2}(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)'(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)\right] \\
&\propto \exp\left[-\frac{1}{2}(\mathbf{Z} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Z} - \mathbf{X}\boldsymbol{\beta}) - \frac{1}{2\sigma^2}(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)'(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)\right] \\
&\propto \exp\left[-\frac{1}{2}\big(\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} - 2\mathbf{Z}'\mathbf{X}\boldsymbol{\beta}\big) - \frac{1}{2\sigma^2}\big(\boldsymbol{\beta}'\boldsymbol{\beta} - 2\boldsymbol{\beta}_0'\mathbf{A}'\boldsymbol{\beta}\big)\right] \\
&(\mathbf{Z}'\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{X}'\mathbf{Z},\ \boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}_0 = \boldsymbol{\beta}_0'\mathbf{A}'\boldsymbol{\beta},\ \text{scalar}) \\
&\propto \exp\left[-\frac{1}{2}\boldsymbol{\beta}'\big(\mathbf{X}'\mathbf{X} + \sigma^{-2}\mathbf{I}_2\big)\boldsymbol{\beta} + \big(\mathbf{Z}'\mathbf{X} + \sigma^{-2}\boldsymbol{\beta}_0'\mathbf{A}'\big)\boldsymbol{\beta}\right]
\end{aligned}$$

Comparing the above with the PDF of a multivariate normal, the variance can easily be seen, which is $(\mathbf{X}'\mathbf{X} + \sigma^{-2}\mathbf{I}_2)^{-1}$. Expanding the PDF of a multivariate normal $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, while also comparing to the scalar case,

$$\begin{aligned}
-\frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\beta} - \boldsymbol{\mu}) = -\frac{1}{2}\boldsymbol{\beta}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\beta} + \underline{\boldsymbol{\mu}'\boldsymbol{\Sigma}^{-1}}\boldsymbol{\beta} - \frac{1}{2}\boldsymbol{\mu}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} \\
\Rightarrow \boldsymbol{\mu}'\boldsymbol{\Sigma}^{-1} = \text{coefficient of } \boldsymbol{\beta} \\
\Rightarrow \boldsymbol{\mu} = (\text{coefficient of } \boldsymbol{\beta} \cdot \boldsymbol{\Sigma})'
\end{aligned}$$

Thus, also considering that covariance matrix and its inverse are always symmetric ($\boldsymbol{\Sigma}' = \boldsymbol{\Sigma}, (\boldsymbol{\Sigma}^{-1})' = (\boldsymbol{\Sigma}^{-1})$), to summarize, we have the following scalar case and matrix case,

$$\begin{aligned}
-\frac{1}{2}a\beta^2 + b\beta \quad & \beta \sim N\left(\frac{b}{a}, \frac{1}{a}\right) \\
-\frac{1}{2}\boldsymbol{\beta}'\mathbf{Q}\boldsymbol{\beta} + \mathbf{r}'\boldsymbol{\beta} \quad & \boldsymbol{\beta} \sim N\big(\mathbf{Q}^{-1}\boldsymbol{r}, \mathbf{Q}^{-1}\big)
\end{aligned}$$

where we can see the mean, which is $(\mathbf{X}'\mathbf{X} + \sigma^{-2}\mathbf{I}_2)^{-1}(\mathbf{X}'\mathbf{Z} + \sigma^{-2}\mathbf{A}\boldsymbol{\beta}_0)$.
Thus,

$$\boxed{\boldsymbol{\beta} \mid \boldsymbol{\beta}_0, \sigma^2, Z, y \sim N\big((\mathbf{X}'\mathbf{X} + \sigma^{-2}\mathbf{I}_2)^{-1}(\mathbf{X}'\mathbf{Z} + \sigma^{-2}\mathbf{A}\boldsymbol{\beta}_0),\ (\mathbf{X}'\mathbf{X} + \sigma^{-2}\mathbf{I}_2)^{-1}\big)}$$

8.2.3 Fully conditional for Z_i

$$\tau(Z_i \mid \boldsymbol{\beta}, \boldsymbol{\beta}_0, \sigma^2, y) \propto \Big[I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0)\Big] \exp\left[-\frac{(Z_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{2}\right]$$

Thus,

$$\boxed{Z_i \mid \boldsymbol{\beta}, y \sim N(\mathbf{x}_i'\boldsymbol{\beta}, 1) \quad \text{truncated to} \begin{cases} (0, \infty), & y_i = 1, \\ (-\infty, 0], & y_i = 0. \end{cases}}$$

8.2.4 Fully conditional for $\boldsymbol{\beta}_0$

$$\begin{aligned} \tau(\boldsymbol{\beta}_0 \mid \boldsymbol{\beta}, \sigma^2, Z, y) &\propto \exp\left[-\frac{1}{2\sigma^2}(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)'(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)\right] \\ &\propto \exp\left[-\frac{1}{2\sigma^2}\big(\boldsymbol{\beta}_0'\mathbf{A}'\mathbf{A}\boldsymbol{\beta}_0 - 2\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}_0\big)\right] \end{aligned}$$

Thus, referring to Bayesian Linear Regression,

$$\boxed{\boldsymbol{\beta}_0 \mid \boldsymbol{\beta}, \sigma^2 \sim N\big((\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\boldsymbol{\beta},\ \sigma^2(\mathbf{A}'\mathbf{A})^{-1}\big)}$$

9 R Document

The R document describing the procedure for Gibbs sampling in this case is given below. The $\mathbf{X}$ and $\mathbf{Y}$ are the same as described in Section 5.

```
library(mvtnorm)

set.seed(1)

X=c(rep(1.69, 6),
    rep(1.72, 6),
    rep(1.76, 6),
    rep(1.78, 6),
    rep(1.81, 6),
    rep(1.84, 6),
    rep(1.86, 6),
```

```
    rep(1.88, 6))
X=cbind(1, as.vector(X))
#X=cbind(1, as.vector(log(X, base=exp(1))))
Y=c(0,0,0,0,0,1,
    0,0,0,1,1,1,
    0,0,0,0,1,1,
    0,0,0,1,1,1,
    0,1,1,1,1,1,
    0,0,1,1,1,1,
    1,1,1,1,1,1,
    1,1,1,1,1,1)

N=length(Y)
p=ncol(X)
A=matrix(c(0, 1), nrow=p, ncol=1)
q=ncol(A)

n_iter=20000
burn_in=5000
thin=10


beta_samples=matrix(0, n_iter, p)
beta0_samples=matrix(0, n_iter, q)
sigma2_samples=rep(0, n_iter)
Z_samples=matrix(0, n_iter, N)

theta_samples=matrix(0, n_iter, p)

XtX=t(X)%*%X
inv_XtX=solve(t(X)%*%X)
inv_AtA=solve(t(A)%*%A)

# Initial values
beta=c(-0.5, 1)
beta0=as.matrix(0)  # q x 1, q < p
sigma2=1
Z=numeric(N)

for (i in 1:n_iter) {

  # --- Sample Z_i (truncated normal) ---
  for (j in 1:N) {
    flag=FALSE
    while (!flag) {
      z_candidate=rnorm(1, mean=X[j,]%*%beta, sd=1)
      if ((Y[j]==1 && z_candidate>0) || (Y[j]==0 && z_candidate<=0)) {
        Z[j]=z_candidate
        flag=TRUE
```

```
      }
    }
  }
  Z_samples[i,]=Z

  # --- Sample beta ---
  # Mean: (XtX + (1/sigma2)*I_2)^{-1}(X'Z + (1/sigma2)*A*beta0)
  # Variance: (XtX + (1/sigma2)*I_2)^{-1}

  sigma_beta=solve(XtX + (1/sigma2)*diag(p))
  mean_beta=sigma_beta %*% (t(X)%*%Z + (1/sigma2)*A %*% beta0)
  beta=as.vector(rmvnorm(1, mean=mean_beta, sigma=sigma_beta))
  beta_samples[i,]=beta

  # --- Sample beta0 ---
  # Mean: (A'A)^{-1}*A*beta
  # Variance: sigma2*(A'A)^{-1}
  mean_beta0=inv_AtA %*% t(A) %*% beta
  sigma_beta0=sigma2*inv_AtA
  beta0=as.vector(rmvnorm(1, mean=mean_beta0, sigma=sigma_beta0))
  beta0_samples[i,]=beta0

  # --- Sample sigma2 ---
  alpha_sigma2=p/2
  beta_sigma2=t(beta - A %*% beta0) %*% (beta - A %*% beta0)/2
  sigma2=1/rgamma(1, shape=alpha_sigma2, rate=beta_sigma2)
  sigma2_samples[i]=sigma2
}

# Remove burn-in and apply thinning
keep=seq(burn_in+1, n_iter, by=thin)
beta_post=beta_samples[keep,]
beta0_post=beta0_samples[keep,]
sigma2_post=sigma2_samples[keep]
Z_post=Z_samples[keep,]

# Coefficients

par(mfrow=c(2, 3), mar=c(4, 4, 2, 1))

# --- Coefficient beta_0 ---
plot(beta_post[,1], type = 'l',
     xlab = "Iteration", ylab = expression(beta[0]),
     main = expression("Trace Plot:" ~ beta[0]))
hist(beta_post[,1], breaks = 25, prob = TRUE,
     xlab = expression(beta[0]),
     main = expression("Posterior:" ~ beta[0]))
lines(density(beta_post[,1]))
acf(beta_post[,1], main = "")
```

```r
title(main = expression("ACF:" ~ beta[0]))

# --- Coefficient beta_1 ---
plot(beta_post[,2], type = 'l',
     xlab = "Iteration", ylab = expression(beta[1]),
     main = expression("Trace Plot:" ~ beta[1]))
hist(beta_post[,2], breaks = 25, prob = TRUE,
     xlab = expression(beta[1]),
     main = expression("Posterior:" ~ beta[1]))
lines(density(beta_post[,2]))
acf(beta_post[,2], main = "")
title(main = expression("ACF:" ~ beta[1]))


# Hyperparameters

par(mfrow=c(2, 3), mar=c(4, 4, 2, 1))

# --- Hyperparameter beta0 ---
plot(beta0_post, type = 'l',
     xlab = "Iteration", ylab = expression(beta[0]),
     main = expression("Trace Plot:" ~ beta[0]))
hist(beta0_post, breaks = 25, prob = TRUE,
     xlab = expression(beta[0]),
     main = expression("Posterior:" ~ beta[0]))
lines(density(beta0_post))
acf(beta0_post, main = "")
title(expression("ACF:" ~ beta[0]))

# --- Hyperparameter sigma2 ---
plot(sigma2_post, type = 'l',
     xlab = "Iteration", ylab = expression(sigma^2),
     main = expression("Trace Plot:" ~ sigma^2))
hist(sigma2_post, breaks = 25, prob = TRUE,
     xlab = expression(sigma^2),
     main = expression("Posterior:" ~ sigma^2))
lines(density(sigma2_post))
acf(sigma2_post, main = "")
title(expression("ACF:" ~ sigma^2))


# Sample statistics
cat("Posterior mean of beta_0:", round(mean(beta_post[,1]), 4), "\n")
cat("Posterior standard deviation of beta_0:", round(sd(beta_post[,1]), 4), "\n")
cat("95% credible interval for beta_0: [",
    round(quantile(beta_post[,1], 0.025), 4), ", ",
    round(quantile(beta_post[,1], 0.975), 4), "]\n\n")
cat("Posterior mean of beta_1:", round(mean(beta_post[,2]), 4), "\n")
```

```r
cat("Posterior standard deviation of beta_1:", round(sd(beta_post[,2]), 4), "\n"
    )
cat("95% credible interval for beta_1: [",
    round(quantile(beta_post[,2], 0.025), 4), ", ",
    round(quantile(beta_post[,2], 0.975), 4), "]\n")
```

10 Results and Diagnostics

Running the above R code, the results will show reasonable trace plots and posterior density plots for the coefficients $\boldsymbol{\beta}$. Plots are shown in Figure 10-2.

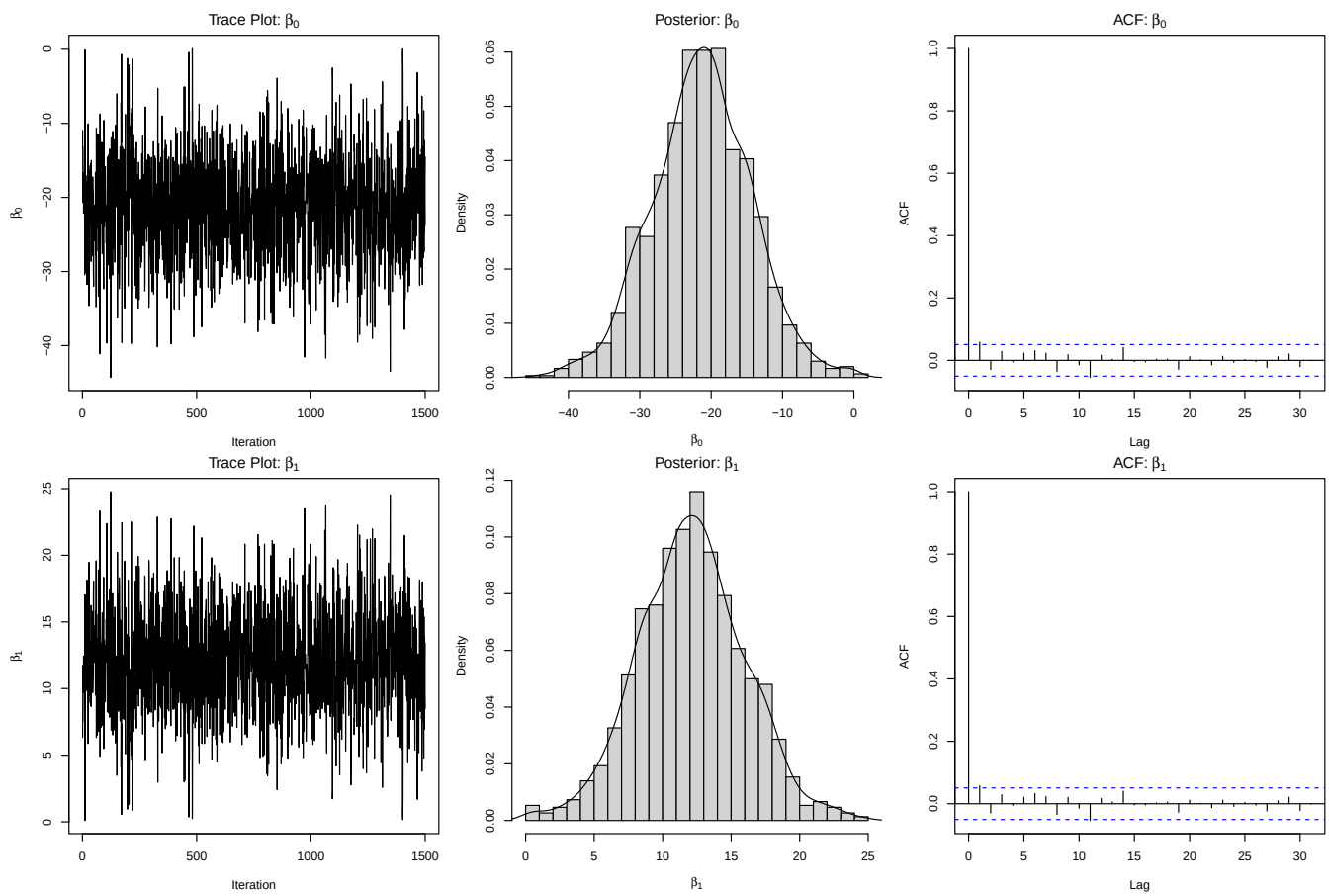

Figure 10-2 Trace plots, posterior density and ACF plots.

After several tryouts, it is found that taking bigger thinning step will help reduce the ACF for the posterior coefficients $\boldsymbol{\beta}$. More iterations and burn-in will help as well. There is randomness involved and for simplicity we have set the seed. We can also report the mean, standard deviation and credible interval from the posterior samples as written in the aforementioned coding.

The posteriors of hyperparameter $\boldsymbol{\beta_0}$ and σ^2 are still very notably flat and diffused. In the trace plots, both parameters exhibit wide-ranging fluctuations across iterations, without settling into a stable or concentrated region as shown in Figure 10-3.

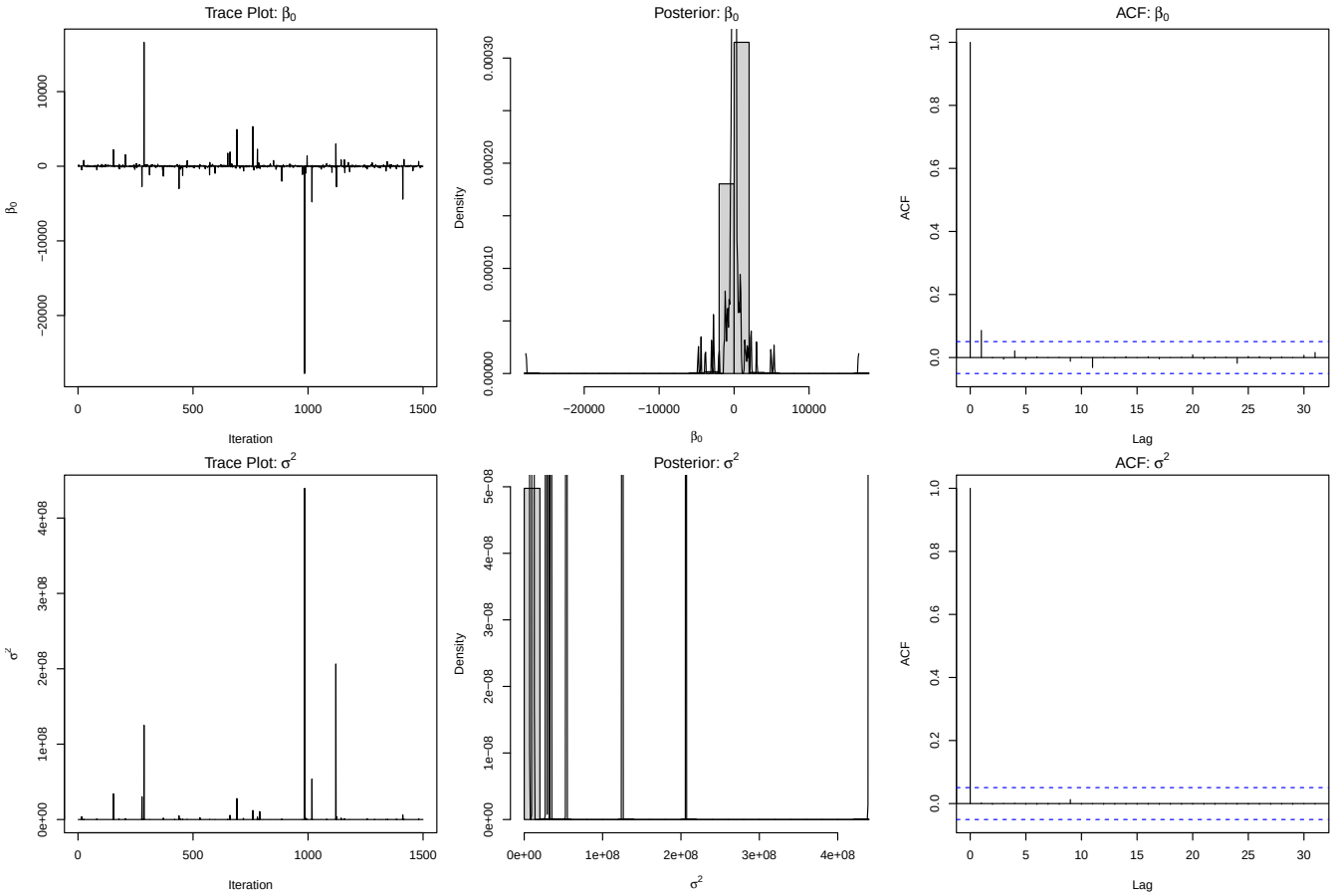

Figure 10-3 Trace plots, posterior density and ACF plots.